Repetition

Loops

for, while, do while

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What and Why?

1 baar hoga

baar baar hoge

```
for (int i=1; i<=3; i++){
    Sout("Shubham");
}
```

4
3
2
1
i

initialization

condition check
work
increment

Output

- Shubham
- Shubham
- Shubham

For Loop

Q/ Print numbers from 1 to 10.

H.W.

~~Ques~~: Print yashika ‘n’ times. Take ‘n’ input from user

How For Loop works : the various parameters

Ques: Print numbers from 1 to 100

Ques: Print all even numbers from 1 to 100

2, 4, 6, 8 . . . . . 98, 100

HW: Print all odd numbers divisible by 3 from 1 to 100

Ques: Print the table of ~~10~~ 17

17 34 51 68 85 102 . . . 170

Ques: Print numbers from ‘n’ to 1.

Decreasing Loop

```
for(int i=n ; i>=1; i--){
    sout(i)
```

3

Ques: Display this AP - 2,5,8,11.. upto 'n' terms

```
for( int i = 2; i<=3n-1 ; i+=3){
    sout(i);
}
```

$$a_n = a + (n-1)d$$

$$2 + (n-1) \cdot 3$$

$$2 + 3n - 3$$

$$\boxed{3n - 1}$$

Ques: Display this GP - 1,2,4,8.. upto 'n' terms

$\boxed{n=5}$

```
int a=1, r=2;
for(int i=1; i<=n; i++){
    sout(a);
    a *= r;
}
```

i: 1 2 3 4 5 6

a: 1 2 4 8 16 32

r: 2

Output

1
2
4
8
16

HW: Print this series - 99,95,91,87,.. upto all terms which are positive

Method-1 Using 'i' condition pata no

Method-2 Using 'a' → no. of terms

Ques: Print all alphabets with their corresponding ASCII values.

A 65

B 66

C 67

.

.

.

.

.

Z 90

HW: Take ‘n’ as input from user and print the following sequence..

1
n
2
n-1
3
n-2
...

n=5

1
5
2
4
3
3
4
2
5
1

Break & continue

is used to skip iterations.

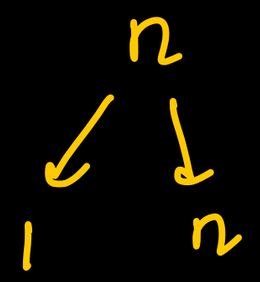
```
for (ini ; condition ; inc/dec){
    if(- - -) break;
}
```

```
for(          ){
}
```

loop → iterations

Ques: WAP to check if a given number is prime or not.

Q. WAP to print if number is composite or not.



2 to n-1 tak koi factor mile 'n' ka to 'n' composite ho jayega

60 → 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60

'i' is factor of 'n' if n%i == 0

Ques: WAP to check if a given number is prime or not.

4 → 1, 2, 4

9 → 1, 3, 9

25 → 1, 5, 25

49 → 1, 7, 49

60 = 1, 2, 3, 4, 5, 6 | 10, 12, 15, 20, 30, 60

$\sqrt{60}$

If 'i' is a factor of 'n' then 'n/i' is also a factor of 'n'

Ques: Print all even numbers from 1 to 100 'Continue Statement'

H.W. Take a number input & print all of its factors.

```
for(int i=1 ; i<=√n ; i++){
    if(n%i==0){
        sout(i);
        sout(n/i);
    }
}
```

While Loop

Usecase : Generally used when conditions are more than one.
Jab bhi iterations nahi pata ho ki kitni hai

```
for(int i=1; i<=10; i++){
|   sout(i);
}
```

```
int i=1;
while(i<=10){
|   sout(i);
|   i++;
}
```

initialization

condition
body
increment

Do-While Loop

Infinite Loop

Ques: Count digits of a number

n = 56132

↓

5613

↓

561

↓

56

↓

5

↓

0

n = 991

↓

99

↓

9

↓

0

```
int n = sc.nextInt();
int count = 0;
while (n!=0){
    n /= 10
    count++;
}
```

3

Ques: Print sum of digits of a number

$n = 5\ 6\ 1\ 4\ 2$

$\text{Sum} = 5+6+1+4+2$

$= 18$

$n = 999$

$\text{sum} = 9+9+9 = 27$

Hint no. 1 .

$5+6+1+4+2 = 2+4+1+6+5$

Hint no. 2

$n \% 10$

Ques: Print sum of digits of a number

n = 5 6 1 4 2 5 6 1 4 5 6 1 5 6 5 0

sum = 0 2 6 7 13 18

n%10
gives the last digit
of any number 'n'

$(-a)\%b = -(a\%b)$

```
while (n != 0){
    sum += n%10
    n /= 10
}
```

~~Ques: Sum of digits of a number~~

int -2^{31} to $2^{31}-1$

long -2^{63} to $2^{63}-1$

Ques: Reverse of a number

$n = 1\ 2\ 8\ 6$

$r = 6\ 8\ 2\ 1$

$6000 + 800 + 20 + 1$

Steps : $6 \rightarrow 60 \rightarrow 68 \rightarrow 680 \rightarrow 682 \rightarrow 6820 \rightarrow \boxed{6821}$

H.W. Print sum of number & its reverse.

Ques: Reverse of a number

n = ~~1 2 8 6~~ ~~128~~ ~~12~~ 1 0

r = ~~0~~ ~~0~~ ~~6~~ ~~60~~ ~~68~~ ~~680~~ ~~682~~ ~~6820~~ 6821

```
while(n != 0){
    r *= 10;
    r += (n%10);
    n /= 10;
}
```

Ques: Factorial of a number

Bhot hi simple.

$\lfloor 5$ or $5! = 5 \times 4 \times 3 \times 2 \times 1$

$\lfloor 8$ or $8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$\lfloor n = 1 \times 2 \times 3 \times \cdots n$

Ques: ‘a’ raise to the power ‘b’

$$a^b = (a \times a \times a \times a \cdots)$$

b times

$$2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$p = 1$

```
for(      ){
  p *= 2
}
```

THANKYOU

Cuties